

Δ_3^1 regularity and failure of definable uniformization

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Abstract

We construct a c.c.c. forcing extension of L , which we assume to have no large cardinals, in which every boldface Δ_3^1 set of reals is Lebesgue measurable and has the Baire property, but there is a Π_2^1 set which cannot be uniformized by any function definable from a real and ordinal parameters. The extension also satisfies $2^{\aleph_0} = 2^{\aleph_1} = \aleph_2$ and Martin's axiom for indestructibly c.c.c. forcings. As a byproduct, it has no wellorder of the reals definable from a real and ordinal parameters; in particular, it has no projectively definable wellorder of the reals.

1 Introduction

We investigate how much projective regularity can coexist with a strong failure of definable uniformization without large-cardinal assumptions. Shelah proved that if every boldface Σ_3^1 set of reals is Lebesgue measurable, then ω_1 is an inaccessible cardinal in L [9, Theorem 5.1]. In contrast, Judah and Shelah showed that, without large-cardinal assumptions, every boldface Δ_3^1 set can simultaneously be Lebesgue measurable and have the Baire property [5, Section 6]; see also [1].

Solovay's model provides a natural motivation for combining this regularity with nonuniformization. The full Lévy-collapse extension of L obtained from an inaccessible cardinal satisfies ZFC, and every set of reals definable from a real and ordinal parameters is Lebesgue measurable and has the Baire property [10, Theorem 2]. At the same time, definable uniformization fails in a strong sense immediately beyond the Σ_2^1 case, for which uniformization is provable in ZFC. Indeed, in this collapse model the lightface Π_2^1 relation

$$R(x, y) \iff y \notin L[x]$$

has total domain but cannot be uniformized by any function definable from a real and ordinal parameters. This is an instance of the homogeneity argument of Lévy [8]; for this precise counterexample, see [7, Section 1].

These results lead to the question addressed in this paper:

Is it consistent relative to ZFC that every boldface Δ_3^1 set of reals is Lebesgue measurable and has the Baire property, while some lightface Π_2^1 relation of total domain has no ordinal-definable uniformization?

The regularity requirement is optimal among the usual full projective pointclasses at the consistency strength of ZFC: replacing Δ_3^1 by Σ_3^1 would already require an inaccessible cardinal in L , independently of the Baire property or any uniformization assertion. In this sense, the question asks whether the strong failure of definable choice in Solovay's model can be retained without large cardinals, with simultaneous measure and category regularity at the highest possible projective level.

We answer the question affirmatively by a c.c.c. forcing extension of L with continuum $\aleph_2$. The counterexample is again the relation $R(x, y) \iff y \notin L[x]$. In fact, the same relation has no uniformizing function definable from any real parameter and finitely many ordinals. The proof combines the Judah–Shelah regularity construction with Lévy’s homogeneity method: we symmetrize the iteration while retaining the bookkeeping needed to replay bounded remainders.

Let $\mathcal{I}$ denote the class of indestructibly c.c.c. forcings. Our main result is the following theorem.

Theorem 1.1. *There is a c.c.c. forcing extension N of L such that:*

- (i) $N \models 2^{\aleph_0} = 2^{\aleph_1} = \aleph_2$;
- (ii) $N \models \text{MA}(\mathcal{I})$;
- (iii) every boldface Δ_3^1 set of reals in N is Lebesgue measurable and has the Baire property;
- (iv) the lightface Π_2^1 relation

$$R(x, y) \iff y \notin L[x]$$

has total domain and has no real-ordinal-definable uniformization in N .

Consequently, if ZFC is consistent, then so is ZFC together with (i)–(iv).

A real-ordinal-definable relation is one definable from a real and finitely many ordinal parameters. The obstruction in part (iv) is thus not confined to any prescribed projective complexity: even arbitrary ordinal parameters together with a real parameter cannot define a uniformizer. In particular, the single lightface Π_2^1 relation R has no ordinal-definable uniformization and no boldface projective uniformization of any complexity; see Corollary 8.5. A further immediate byproduct is that N has no wellorder of the reals definable from a real and ordinal parameters. Since the construction is c.c.c. over L , it also preserves ω_1^L . Thus the proof yields, from $\text{Con}(\text{ZFC})$, the consistency of

$$\text{MA}(\mathcal{I}) + \omega_1^L = \omega_1 + \text{“there is no real-ordinal-definable wellorder of } \mathbb{R}\text{.”}$$

This is stated explicitly in Corollary 8.6.

This byproduct is also related to a question of Caicedo and Friedman. They proved that

$$\text{BPFA} + \omega_1^L = \omega_1$$

implies the existence of a lightface Σ_3^1 wellorder of the reals, and asked whether

$$\text{MA} + \omega_1^L = \omega_1$$

is consistent with the nonexistence of a projective wellorder of the reals [2, Section 4, Question 2]. The present theorem does not answer their question, since $\text{MA}(\mathcal{I})$ is weaker than full Martin’s axiom. It does, however, give a restricted-forcing-axiom analogue with a stronger anti-definability conclusion: $\text{MA}(\mathcal{I}) + \omega_1^L = \omega_1$ is consistent with the nonexistence of any real-ordinal-definable wellorder of the reals. Thus the symmetry used to obtain nonuniformization simultaneously yields a natural partial result in the direction of the Caicedo–Friedman question.

There is a further reason why this question is sensitive to the way in which Martin’s axiom is forced. The construction in [3] implicitly shows that one can prepare a universe W with the following property. Fix the canonical version of the standard Jech-style finite-support proof of MA_{ω_1} : relevant names are ordered by a canonical ground-model wellorder and

scheduled by a fixed canonical pairing $\omega_2 \times \omega_2 \cong \omega_2$. If $\mathbb{P}$ is the resulting ω_2 -stage iteration over W , then

$$W \models \mathbb{P} \Vdash \text{“there is a lightface } \Delta_3^1 \text{ wellorder of the reals.”}$$

Thus every generic extension obtained from this canonically prescribed version of the usual $\mathbf{MA}_{\omega_1}$ construction acquires such a wellorder automatically. The point implicit in [3, Section 1, especially the discussion preceding Lemma 1.28] is that the prepared system of Suslin-tree blocks and projectively recognizable T -models makes the canonical stage-and-name ordering projectively recognizable. This produces the lightface Δ_3^1 wellorder. Hence merely applying the standard canonical bookkeeping can itself introduce a low-complexity wellorder; avoiding this symmetry breaking is an additional motivation for the equivariant bookkeeping used here. We stress that this observation is implicit in the construction of [3], rather than stated there as a separate theorem.

The construction has three components. First, a scheduled forcing $\dot{\mathbb{Q}}$ is replaced by a finite-support product of countably many copies of all its translates under a coherent automorphism group. Permuting identical copies makes the product weakly homogeneous, even when $\dot{\mathbb{Q}}$ itself has no useful homogeneity. Taking all translates allows the automorphisms of the preceding iteration to lift. At every stage below ω_2 , the group and a dense subset of the initial forcing have size at most $\aleph_1$.

Second, one designated factor at each stage almost-disjointly codes a subset of ω_1 determining the preceding generic filter. If r is its coding real, the extension by that factor is exactly $L[r]$. The quotient of the full iteration over this intermediate model remains weakly homogeneous. Consequently, every real parameter in the final model belongs to some such $L[r]$. A uniformizer of R definable from that parameter and ordinals would select at r a real which, by homogeneity, belongs to $L[r]$, contrary to the definition of R .

Third, the bookkeeping supplies generic filters for all earlier orders in $\mathcal{I}$ of size at most $\aleph_1$, including their cones. This both yields $\mathbf{MA}(\mathcal{I})$ and permits the Judah–Shelah regularity argument: a bounded remainder evaluated over a random or Cohen real can be forced with again below a prescribed condition. Almost-disjoint coding is used only to reconstruct intermediate models from particular coding reals, so it imposes no additional restriction on this bookkeeping.

The symmetrization used below is related to Shelah’s amalgamation method, which extends isomorphisms of complete subalgebras by enlarging the ambient forcing [9, Claim 7.13]. A more direct precursor for the successor construction is the use of automorphism-invariant families of factors in the self-indexed products of Goldstern, Kellner, Mejía, and Shelah [4, Section 4.C, especially Fact 4.13]. Here we use an explicitly orbit-indexed presentation and countably many repetitions of each factor. The repetitions provide weak homogeneity even after fixing a designated coordinate. Combined with almost-disjoint coding of initial generic filters, this yields the homogeneous quotients over models $L[r]$ needed for the nonuniformization argument.

Section 2 records the closure properties of $\mathcal{I}$. Sections 3 and 4 develop real coding and relative homogeneity. The iteration and its bookkeeping are given in Section 5. After establishing the required measure and category covering properties in Section 6, we prove regularity in Section 7 and the failure of definable uniformization in Section 8.

2 Conventions and the forcing class

We identify the space of reals with 2^ω when discussing measure or category; recursive codings transfer the assertions to ω^ω . Lightface pointclasses have no real parameters and boldface

pointclasses allow them. A uniformization of $A \subseteq \mathbb{R}^2$ is a subset $F \subseteq A$ which is the graph of a function on $\text{dom}(A)$; thus F contains exactly one point of each nonempty vertical section of A .

All forcing orders have a largest condition, and smaller conditions are stronger. Dense suborders and Boolean completions may be used when factoring forcings. A forcing from an earlier model is always interpreted as the *fixed earlier order*, unless recomputation is explicitly specified. Standard forcing facts used below may be found in [6].

Definition 2.1. A forcing $\mathbb{Q}$ belongs to $\mathcal{I}$ if $\mathbb{Q} \times \mathbb{S}$ is c.c.c. for every c.c.c. forcing $\mathbb{S}$. Equivalently, $\mathbb{Q}$ is c.c.c. and remains c.c.c. after forcing with any c.c.c. order. The statement $\text{MA}(\mathcal{I})$ requires a filter meeting any family of fewer than $2^{\aleph_0}$ dense subsets of any member of $\mathcal{I}$.

Lemma 2.2. *If $\mathbb{Q} \in \mathcal{I}$ and $\mathbb{S}$ is c.c.c., then*

$$\Vdash_{\mathbb{S}} \check{\mathbb{Q}} \in \mathcal{I}.$$

Moreover, if $\mathbb{T} \in \mathcal{I}$, forcing with $\mathbb{T}$ preserves nonmembership in $\mathcal{I}$ of every fixed ground-model order.

Proof. Suppose $\mathbb{S}$ forces that $\dot{\mathbb{U}}$ is c.c.c. Then $\mathbb{S} * \dot{\mathbb{U}}$ is c.c.c., so $\mathbb{Q} \times (\mathbb{S} * \dot{\mathbb{U}})$ is c.c.c. Reordering the fixed $\mathbb{Q}$ factor shows that $\mathbb{S}$ forces $\check{\mathbb{Q}} \times \dot{\mathbb{U}}$ to be c.c.c. This proves the first assertion.

For the second, an uncountable antichain in a fixed order remains one in any c.c.c. extension. Otherwise, nonmembership of a ground-model $\mathbb{Q}$ in $\mathcal{I}$ is witnessed by a ground-model c.c.c. order $\mathbb{S}$ and an uncountable antichain in $\mathbb{Q} \times \mathbb{S}$. Forcing with $\mathbb{T} \in \mathcal{I}$ preserves the c.c.c. of $\mathbb{S}$, preserves ω_1 , and does not change compatibility between elements of the two fixed orders. The same witness therefore still proves $\mathbb{Q} \notin \mathcal{I}$. $\square$

Lemma 2.3. *The class $\mathcal{I}$ is closed under cones, finite-support products, and finite-support iterations whose iterands are forced to belong to $\mathcal{I}$. Every tail of such an iteration belongs to $\mathcal{I}$ in its appropriate intermediate universe.*

Proof. Cones inherit the defining product property. Finite products do as well. For an arbitrary finite-support product, apply a Δ -system argument to an uncountable family of finite supports. On the common finite root, the corresponding finite product is c.c.c.; compatible root restrictions can be combined with the disjoint parts of the supports. This argument also works after any c.c.c. extension, by Lemma 2.2.

For the successor step of an iteration, suppose $\mathbb{P} \in \mathcal{I}$ and $\mathbb{P} \Vdash \dot{\mathbb{Q}} \in \mathcal{I}$. Let $\mathbb{S}$ be c.c.c. In the $\mathbb{P}$ -extension, $\mathbb{S}$ is still c.c.c., so $\dot{\mathbb{Q}}$ remains c.c.c. after forcing with $\mathbb{S}$. Commuting the fixed $\mathbb{S}$ factor gives that $\mathbb{S} * (\mathbb{P} * \dot{\mathbb{Q}})$ is c.c.c. Thus $\mathbb{P} * \dot{\mathbb{Q}} \in \mathcal{I}$. At a limit, carry out the usual finite-support c.c.c. iteration argument in an arbitrary c.c.c. extension; the successor observation verifies its hypotheses. This proves the assertion by induction. The same induction performed in an intermediate extension proves the assertion about tails. $\square$

Every Knaster forcing belongs to $\mathcal{I}$: from an uncountable family of pairs, first thin on its Knaster coordinate to a pairwise compatible family and then use the c.c.c. of the other coordinate. In particular, every σ -linked forcing belongs to $\mathcal{I}$. Cohen forcing, random forcing, and the almost-disjoint forcing defined below all belong to $\mathcal{I}$. For random forcing, the usual countable linked cover is obtained by requiring a condition to occupy more than half of a specified nonempty basic clopen set; every positive Borel condition satisfies such a requirement after choosing a suitable basic clopen set.

Lemma 2.4. *Let $\mathbb{Q} \in \mathcal{I}$ and let $\langle D_\xi : \xi < \omega_1 \rangle$ be dense subsets of $\mathbb{Q}$. There is a suborder $\mathbb{Q}_0$ of size at most $\aleph_1$ such that $\mathbb{Q}_0 \in \mathcal{I}$, each $D_\xi \cap \mathbb{Q}_0$ is dense in $\mathbb{Q}_0$, and a filter on $\mathbb{Q}_0$ generates a filter on $\mathbb{Q}$.*

Proof. Choose a subset X of $\mathbb{Q}$ containing its largest condition and of size at most $\aleph_1$, and close it under two families of choices. For every compatible pair in X , add a common extension in $\mathbb{Q}$. For every $p \in X$ and $\xi < \omega_1$, add an extension of p in D_ξ . Repeat this construction ω times, and let $\mathbb{Q}_0$ have the inherited order on the union. Compatibility of two elements of $\mathbb{Q}_0$ is the same in $\mathbb{Q}_0$ and $\mathbb{Q}$. Hence any antichain in $\mathbb{Q}_0 \times \mathbb{S}$ is an antichain in $\mathbb{Q} \times \mathbb{S}$. This proves $\mathbb{Q}_0 \in \mathcal{I}$. Density follows from the second closure requirement, and upward closure in $\mathbb{Q}$ of a directed filter on $\mathbb{Q}_0$ gives the last assertion. No complete-suborder claim is needed. $\square$

3 Capturing an initial extension by a real

Fix in L a canonical almost-disjoint family

$$\vec{d} = \langle d_\xi : \xi < \omega_1 \rangle \subseteq [\omega]^\omega.$$

All forcings in the construction are c.c.c., so this remains the same ω_1 -indexed family throughout the iteration.

Definition 3.1. For $A \subseteq \omega_1$, a condition of $\mathbb{A}(A)$ is a pair (s, F) , where $s \in 2^{<\omega}$ and F is a finite subset of A . We put $(t, F') \leq (s, F)$ if $s \subseteq t$, $F \subseteq F'$, and

$$n \in [|s|, |t|), \quad t(n) = 1 \quad \implies \quad n \notin \bigcup_{\xi \in F} d_\xi.$$

Lemma 3.2. *The forcing $\mathbb{A}(A)$ is σ -centered. If H is generic over M and $r = \bigcup \{s : (s, F) \in H\}$, then*

$$\xi \in A \iff \{n : r(n) = 1\} \cap d_\xi \text{ is finite.} \quad (1)$$

Furthermore H is definable from r and the ground-model order, and $M[H] = M[r]$.

Proof. Conditions with the same stem form a centered set, since their finite side conditions can be united. Stems can be made arbitrarily long. For $\xi \in A$, the requirement $\xi \in F$ is dense and ensures finite intersection. For $\xi \notin A$, every condition can add a new 1 on d_ξ beyond any specified bound: only finitely many almost-disjoint sets are forbidden by its side condition. These dense requirements give (1).

More precisely, for a ground-model condition (s, F) ,

$$(s, F) \in H \iff s \subseteq r \text{ and } \forall \xi \in F \forall n \geq |s| (n \in d_\xi \implies r(n) = 0). \quad (2)$$

The forward implication follows from the order. For the reverse implication, a condition satisfying the right side is compatible with every element of H : its stem and the stem of an element of H are comparable initial segments of r , and the longer stem obeys both finite side conditions. Genericity for the dense set of conditions extending or incompatible with (s, F) then puts (s, F) into H . Equation (2) proves the assertion about the extension. $\square$

Lemma 3.3. *Suppose $\mathbb{P} \in L$ is c.c.c. and has a dense subset $D \in L$ of size at most $\aleph_1$. Let G be $\mathbb{P}$ -generic over L and $M = L[G]$. There is $A \subseteq \omega_1$ in M such that, if $H \subseteq \mathbb{A}(A)$ is generic over M with coding real r , then*

$$M[H] = L[r].$$

Proof. Fix in L a surjection $e : \omega_1 \rightarrow D$, and put

$$A = \{\xi < \omega_1 : e(\xi) \in G\}.$$

By (1), r determines A using the fixed family $\vec{d} \in L$. From A and e , it determines $G \cap D$, and hence G . Thus $L[G] \subseteq L[r]$. Lemma 3.2 gives $M[H] = M[r]$, which is consequently $L[r]$. The reverse inclusion also follows directly from $r \in M[H]$ and the minimality of $L[r]$. $\square$

The reconstruction in Lemma 3.3 uses the particular real added by the specified factor. Formula (1) does not require uniqueness of a real coding A , and the same almost-disjoint family may be used to code other subsets of ω_1 .

4 Symmetrizing the iteration

4.1 A successor step

A group $\Gamma \leq \text{Aut}(\mathbb{P})$ witnesses *weak homogeneity* if for all $p, q \in \mathbb{P}$ some $h \in \Gamma$ makes $h(p)$ compatible with q . The action on names is the usual recursively induced action.

When a forcing has a natural presentation, we retain a name for an isomorphism between that presentation and its normalized presentation, together with any associated generic real. Automorphisms transport this entire package.

Lemma 4.1. *Let $\mathbb{P}$ be a forcing which preserves CH and has a dense subset D of size at most $\aleph_1$. Let $\Gamma \leq \text{Aut}(\mathbb{P})$ have size at most $\aleph_1$, let $k < \omega$, and let $\langle \dot{Q}_i : i < k \rangle$ be $\mathbb{P}$ -names such that, for every $i < k$, $\mathbb{P}$ forces that $\dot{Q}_i \in \mathcal{I}$ and that $\dot{Q}_i$ is an order on a subset of the ground-model ordinal ω_1 , with 0 as its largest condition. Put $J = k \times \Gamma \times \omega$ and define the $\mathbb{P}$ -name*

$$\dot{S} = \prod_{(i,g,n) \in J}^{\text{FS}} g(\dot{Q}_i). \quad (3)$$

Then the following hold:

- (i) $\mathbb{P} \Vdash \dot{S} \in \mathcal{I}$, and the two-step iteration $\mathbb{P} * \dot{S}$ has a dense subset of size at most $\aleph_1$ in the ground model.
- (ii) There are a group $\Gamma^+ \leq \text{Aut}(\mathbb{P} * \dot{S})$ of size at most $\aleph_1$ and an injective homomorphism $h \mapsto \hat{h}$ from Γ into Γ^+ such that $\hat{h}$ restricts to h on the canonical copy of $\mathbb{P}$ in $\mathbb{P} * \dot{S}$.
- (iii) If Γ witnesses weak homogeneity of $\mathbb{P}$, then Γ^+ witnesses weak homogeneity of $\mathbb{P} * \dot{S}$.

Proof. The coordinates (i, g, n) and (i, g, n') carry identical copies of the same name. The index n is a copy index, not a stage of the outer iteration. Since automorphisms preserve the forcing relation, every factor in (3) is forced to belong to $\mathcal{I}$. Lemma 2.3 gives the assertion about $\dot{S}$.

There are two types of automorphisms. First, for $h \in \Gamma$, lift h by its old action on $\mathbb{P}$, by transport of names, and by the permutation

$$(i, g, n) \mapsto (i, hg, n). \quad (4)$$

This is valid because $h(g(\dot{Q}_i)) = (hg)(\dot{Q}_i)$. The lifts respect composition and are injective on the old automorphism group.

Second, let

$$K = \bigoplus_{(i,g) \in k \times \Gamma} \text{Sym}_{\text{fin}}(\omega)$$

act by finitely supported permutations of the copy index in each fiber. These automorphisms fix $\mathbb{P}$ pointwise. Define Γ^+ to be the group generated by the lifts and these permutations. There are at most $\aleph_1$ generators, and hence at most $\aleph_1$ group elements.

For the density assertion, let $\text{Fn}(J, \omega_1)$ denote the set of finite partial functions from J to the ground-model ordinal ω_1 . Define, in the ground model,

$$D^+ = \{(p, \check{f}) : p \in D, f \in \text{Fn}(J, \omega_1), p \Vdash \check{f} \in \dot{\mathbb{S}}\}.$$

This set is dense in $\mathbb{P} * \dot{\mathbb{S}}$. Indeed, given a condition $(p, \dot{s})$, first strengthen p to decide the finite support of $\dot{s}$ and all its ordinal labels. Thus some $p' \leq p$ forces $\dot{s} = \check{f}$ for a ground-model $f \in \text{Fn}(J, \omega_1)$. Choose $d \in D$ with $d \leq p'$. Then $(d, \check{f}) \in D^+$ and $(d, \check{f}) \leq (p, \dot{s})$. Since $|D|, |J| \leq \aleph_1$, we have $|D^+| \leq \aleph_1$.

Finally, suppose that Γ witnesses weak homogeneity of $\mathbb{P}$. It suffices to consider two conditions $(p, \check{f}), (q, \check{u}) \in D^+$. Choose $h \in \Gamma$ and $r \leq h(p), q$. Put

$$f^h = \{((i, hg, n), f(i, g, n)) : (i, g, n) \in \text{dom}(f)\},$$

so that $\widehat{h}(p, \check{f}) = (h(p), \check{f}^h)$. For each $(i, g) \in k \times \Gamma$, let

$$A_{i,g} = \{n : (i, g, n) \in \text{dom}(f^h)\}, \quad B_{i,g} = \{n : (i, g, n) \in \text{dom}(u)\}.$$

Choose $\pi_{i,g} \in \text{Sym}_{\text{fin}}(\omega)$ such that

$$\pi_{i,g}[A_{i,g}] \cap B_{i,g} = \emptyset, \quad A_{i,g} \cap B_{i,g} = \emptyset \implies \pi_{i,g} = \text{id}_\omega.$$

Since only finitely many $A_{i,g}$ are nonempty, $\pi = (\pi_{i,g})_{(i,g) \in k \times \Gamma} \in K$. Let $v = \pi \cdot f^h$, that is,

$$v(i, g, \pi_{i,g}(n)) = f^h(i, g, n) \quad ((i, g, n) \in \text{dom}(f^h)).$$

Then $\text{dom}(v) \cap \text{dom}(u) = \emptyset$, and

$$(r, v \check{\cup} u) \leq (\pi \circ \widehat{h})(p, \check{f}), (q, \check{u}).$$

Thus $\pi \circ \widehat{h} \in \Gamma^+$ witnesses the required compatibility. $\square$

4.2 Coherent groups and limits

We apply Lemma 4.1 at every stage of a finite-support iteration. Write its groups as Γ_α , identify every element of Γ_α with its canonical lift at later stages, and take direct limits of the groups at limit ordinals. Every automorphism preserves each earlier outer iteration stage.

At a limit λ , conditions have finite outer support and so belong to an earlier initial forcing. Consequently weak homogeneity passes to $\mathbb{P}_\lambda$. The determined dense presentations are coherent, and their union is dense at a limit. If $\lambda < \omega_2$, the union of $|\lambda| \leq \aleph_1$ sets of size at most $\aleph_1$ still has size at most $\aleph_1$. The same calculation applies to the groups.

4.3 Fixing a single coordinate

Global weak homogeneity is not enough to rule out real-parameter definable uniformization. We need the following relative statement. It is the reason for using separate fibers for the different kinds of scheduled forcing, rather than first bundling them into one factor.

Lemma 4.2. *Consider a coherently symmetrized iteration as above. Fix a stage α and one coordinate $j = (i, g, n)$ of $\dot{S}_\alpha$. The forcing*

$$\mathbb{P}_\alpha * g(\dot{Q}_{\alpha,i})$$

embeds completely in the iteration through the designated coordinate j . Over its generic extension, the quotient of the full iteration is weakly homogeneous.

Proof. Let M be the $\mathbb{P}_\alpha$ -extension and let H_j be the generic filter at the designated factor. At stage α , the remaining forcing is the finite-support product of all other factors. In the fiber containing j , countably infinitely many copies remain; every other fiber is unchanged. The subgroup of K_α fixing j pointwise witnesses weak homogeneity of this remaining product by the disjoint-support argument. It fixes $\mathbb{P}_\alpha$ and the entire j -factor pointwise.

Inductively, lift these automorphisms to every later stage and adjoin the new copy permutations there. All these automorphisms continue to fix $\mathbb{P}_\alpha * g(\dot{Q}_{\alpha,i})$ pointwise. Thus they induce automorphisms of the quotient after its generic filter has been fixed. At a successor stage, apply exactly the proof of Lemma 4.1, now to quotient conditions. The inductive group first makes the old quotient coordinates compatible; new copy permutations make the two new finite supports disjoint. The later orbit product contains the translates by every member of the full old group, hence in particular by each member of this subgroup.

At limits, quotient conditions still have finite outer support. The groups and the compatibility argument therefore pass to the direct limit. This proves weak homogeneity of the entire quotient over $M[H_j]$. Complete embeddability follows initially from the product factorization and then from the complete initial-segment embeddings of the iteration. $\square$

Remark 4.3. The proof does not require the fixed factor to be homogeneous. For example, that factor can be $\mathbb{A}(A)$ or an arbitrary scheduled member of $\mathcal{I}$. Nor does it require two different data-dependent coding forcings to be isomorphic. It permutes identical copies and transports whole names under actual automorphisms of the preceding forcing.

5 The iteration and its bookkeeping

Work in L , and put $\kappa = \omega_2$. We recursively construct

$$\langle \mathbb{P}_\alpha, \dot{S}_\alpha, \Gamma_\alpha : \alpha < \kappa \rangle$$

with finite support. Initially $\mathbb{P}_0$ and Γ_0 are trivial. At stage α , use four forcing packages in (3):

- (i) $\dot{Q}_{\alpha,0}$ is Cohen forcing;
- (ii) $\dot{Q}_{\alpha,1}$ is random forcing as computed in the $\mathbb{P}_\alpha$ -extension;
- (iii) $\dot{Q}_{\alpha,2}$ is $\mathbb{A}(\dot{A}_\alpha)$, where $\dot{A}_\alpha$ codes the $\mathbb{P}_\alpha$ -generic filter as in Lemma 3.3;
- (iv) $\dot{Q}_{\alpha,3}$ is the current bookkeeping request, provided it is an order of size at most $\aleph_1$ belonging to $\mathcal{I}$ in the current extension; otherwise it is trivial.

Each package is normalized as in the previous section. Set

$$\dot{S}_\alpha = \prod_{(i,g,n) \in 4 \times \Gamma_\alpha \times \omega}^{\text{FS}} g(\dot{Q}_{\alpha,i}), \quad \mathbb{P}_{\alpha+1} = \mathbb{P}_\alpha * \dot{S}_\alpha.$$

Use the group and its canonical lift described there. We refer to the factor with indices $(i, \text{id}, 0)$ as the designated factor of type i at stage α .

5.1 Enumeration of requests

At each stage β , choose a determined dense subset $D_\beta \in L$ of size at most $\aleph_1$ and an enumeration of it in L . A name for a subset of ω_1 can be represented by a nice name using this dense subset. In L , the number of these names is at most

$$(\aleph_1^{\aleph_0})^{\aleph_1} = 2^{\aleph_1} = \aleph_2. \quad (5)$$

An order on a subset of ω_1 , its domain and its largest condition can all be coded by one subset of ω_1 . Enumerate these names as $\langle \tau_{\beta,\xi} : \xi < \kappa \rangle$, with repetitions if necessary.

Reserve cofinally many slots above β for each pair $(\beta, \xi) \in \kappa \times \kappa$. Such a reservation is possible since $\kappa^2 = \kappa$: partition κ into κ many sets of size κ , and discard from the set assigned to (β, ξ) its initial part below $\beta + 1$. At a reserved slot, read the old name as a name over the current forcing and perform the test in (iv). Unused slots may carry the trivial request. Membership in $\mathcal{I}$ is tested in the current intermediate model.

Proposition 5.1. *The construction is well-defined. For every $\alpha < \kappa$, $\mathbb{P}_\alpha$ has a dense subset of size at most $\aleph_1$, forces CH, and belongs to $\mathcal{I}$. Its group Γ_α has size at most $\aleph_1$ and witnesses weak homogeneity. The final forcing belongs to $\mathcal{I}$, has a dense subset of size κ , preserves cardinals, and forces*

$$2^{\aleph_0} = 2^{\aleph_1} = \aleph_2.$$

Proof. Induct on α . A c.c.c. forcing with a dense subset of size $\aleph_1$ over L preserves CH, since the number of nice names for reals is at most $(\aleph_1^{\aleph_0})^L = \aleph_1$. Consequently full random forcing at a proper stage has size $\aleph_1$. The almost-disjoint forcing and each normalized bookkeeping request also have size at most $\aleph_1$ and belong to $\mathcal{I}$. Lemmas 4.1 and 2.3 give the successor step. The observations about limits in the previous section give the limit step, including the bounds on the groups. Thus all four inputs at every stage have the required size and chain condition.

At κ , the union of the determined dense presentations has size at most κ , and the final forcing is c.c.c. and belongs to $\mathcal{I}$. In L , $\kappa^{\aleph_0} = \kappa^{\aleph_1} = \kappa$. Counting nice names therefore gives upper bounds κ for both $2^{\aleph_0}$ and $2^{\aleph_1}$ in the extension. The designated Cohen coordinate at each stage adds a real not in the preceding intermediate model. These κ reals are pairwise distinct. This proves both equalities and also that the density of the final forcing is κ . Cardinal preservation follows from the c.c.c. $\square$

Fix henceforth a generic $G \subseteq \mathbb{P}_\kappa$, and write

$$M_\alpha = L[G_\alpha], \quad N = L[G].$$

Every real of N belongs to some M_α . More generally, every subset of ω_1 in N belongs to some M_α : a nice name uses at most $\aleph_1$ many finite-support conditions, whose outer supports are bounded in κ .

Lemma 5.2. *Suppose $\mathbb{Q} \in M_\beta$, $M_\beta \models \mathbb{Q} \in \mathcal{I}$, and $M_\beta \models |\mathbb{Q}| \leq \aleph_1$. For every $q \in \mathbb{Q}$ and arbitrarily large $\delta < \kappa$, the construction supplies a filter on $\mathbb{Q}$ containing q which is generic over M_δ . In particular, it is generic over M_β .*

Proof. Normalize the cone $\mathbb{Q} \restriction q$ as an order on a subset of ω_1 . Its code is the evaluation of one of the names $\tau_{\beta,\xi}$, after increasing β if necessary to include the chosen normalization. By Lemma 2.2, this fixed order remains in $\mathcal{I}$ in every later intermediate model. At each of its cofinally many reserved slots, the designated request factor therefore forces with that order. Its coordinate filter is generic over the model just before that block, since it is one

factor of a genuine product. Transfer the filter through the normalization and take upward closure in $\mathbb{Q}$. The resulting filter contains q . We have forced below q , rather than assumed that a generic for the whole order happens to contain q . $\square$

Proposition 5.3. *The model N satisfies $\text{MA}(\mathcal{I})$.*

Proof. Let $\mathbb{Q} \in \mathcal{I}^N$ and let $\langle D_\xi : \xi < \omega_1 \rangle$ be dense subsets of $\mathbb{Q}$ in N . By Lemma 2.4, reduce to an order $\mathbb{Q}_0$ of size at most $\aleph_1$ whose compatibility is inherited from $\mathbb{Q}$. Normalize it on a subset of ω_1 , and code its order and the family $\langle D_\xi \cap \mathbb{Q}_0 : \xi < \omega_1 \rangle$ by a subset of ω_1 . This normalized data belongs to some M_β . In the rest of the proof $\mathbb{Q}_0$ denotes the normalized order; the map back to the original order is used only in N .

The order $\mathbb{Q}_0$ already belongs to $\mathcal{I}$ in M_β . Otherwise Lemma 2.2, applied to the $\mathcal{I}$ tail from M_β to N , preserves a witness to $\mathbb{Q}_0 \notin \mathcal{I}$, contradicting its membership in $\mathcal{I}^N$. The recorded dense sets are dense already in M_β , since the order and each of these sets are fixed there. Lemma 5.2 supplies a filter meeting them all. It generates a filter on $\mathbb{Q}$ with the required property. Since $2^{\aleph_0} = \aleph_2$, families of size at most $\aleph_1$ are precisely the relevant families in $\text{MA}(\mathcal{I})$. $\square$

6 Largeness of the sets of earlier generics

We include the covering argument needed to pass from generic reals to ordinary Lebesgue measurability and the Baire property.

Lemma 6.1. *In a model of $\text{MA}(\mathcal{I})$ with continuum $\aleph_2$, every union of at most $\aleph_1$ null sets is null, and every union of at most $\aleph_1$ meager sets is meager.*

Proof. For measure, fix $0 < \epsilon < 1$ rational and use the order

$$\mathbb{A}_\epsilon = \{U \subseteq 2^\omega : U \text{ open and } \mu(U) < \epsilon\}, \quad V \leq U \iff V \supseteq U.$$

This order is σ -linked. For a clopen set C and a positive rational δ with $\mu(C) + 2\delta < \epsilon$, the class

$$\{U : C \subseteq U, \mu(U \setminus C) < \delta\}$$

is linked, since the union of two of its members still has measure less than ϵ . These countably many classes cover $\mathbb{A}_\epsilon$, by clopen approximation from inside an open set. Thus $\mathbb{A}_\epsilon \in \mathcal{I}$.

If $\langle Z_\xi : \xi < \omega_1 \rangle$ is a family of null sets, then $\{U : Z_\xi \subseteq U\}$ is dense for each ξ : add a small enough open cover of Z_ξ to the current condition. A filter meeting these dense sets gives an open union O_ϵ containing every Z_ξ , with $\mu(O_\epsilon) \leq \epsilon$. For the measure bound, every finite union of members of the filter is contained in a common stronger condition; second countability then gives the bound for their full union. Repeating this for $\epsilon = 2^{-n-1}$ yields a Borel null set containing $\bigcup_\xi Z_\xi$.

For category, first write the given meager sets as subsets of a family $\langle F_\xi : \xi < \omega_1 \rangle$ of closed nowhere dense sets. Consider conditions (s, E) where s is a finite function on ω , each $s(n)$ is a finite family of basic clopen sets, and $E \subseteq \omega_1 \times \omega$ is finite, subject to

$$(\xi, n) \in E \implies F_\xi \cap \bigcup s(n) = \emptyset.$$

A missing value of s is read as empty. Extension adds clopen sets to the finite values of s and adds pairs to E . Conditions with the same s form a centered set: unite their finite sets E . There are only countably many s , so this forcing is σ -centered and belongs to $\mathcal{I}$.

For each ξ , the requirement that $(\xi, n) \in E$ for some n is dense; choose a new n . For each n and each nonempty basic clopen B , the requirement that $s(n)$ contain a nonempty basic

clopen subset of B is dense. Indeed, only finitely many nowhere dense sets are currently assigned to n . There are $\aleph_1$ many requirements altogether. A filter meeting them produces open dense sets

$$O_n = \bigcup \{ \bigcup s(n) : (s, E) \in H \}.$$

If (ξ, n) belongs to the ledger of some condition in the filter, directedness implies $F_\xi \cap O_n = \emptyset$. Hence the closed nowhere dense sets $2^\omega \setminus O_n$ cover all the F_ξ . Their union is a meager cover as required. $\square$

Corollary 6.2. *For each $\alpha < \kappa$, the set of reals in N which are random over M_α is conull, and the set of reals which are Cohen over M_α is comeager.*

Proof. There are only $\aleph_1$ real codes in M_α . A real is random over M_α exactly when it avoids every Borel null set coded there. The full interpretations in N of these Borel codes are still null. Lemma 6.1 makes their union null. Similarly, the nongeneric Cohen reals form the union of the Borel meager sets coded in M_α , or equivalently can be covered using its closed nowhere dense codes. This union is meager. $\square$

7 The Judah–Shelah transfer and replay argument

7.1 Fixed-order transfer

Lemma 7.1. *Suppose $M \models q \Vdash_{\mathbb{Q}} \theta(\dot{z}, a)$, where θ is Π_2^1 , a is a real of M , and $\mathbb{Q}$, q and $\dot{z}$ belong to M . In any forcing extension $M[K]$, the same assertion holds for the fixed order $\mathbb{Q}$ and the same name.*

Proof. In a product extension by $\mathbb{Q}$ and the forcing producing K , the $\mathbb{Q}$ -generic filter through q first yields $\theta(\dot{z}^H, a)$ over M . Shoenfield absoluteness preserves this assertion when the other generic is added. The product can be taken in the opposite order, which proves that $M[K]$ sees the same forcing assertion. This is the usual Π_2^1 transfer used in the Judah–Shelah argument. $\square$

In the application below, the remainder is first evaluated over a generic real. The resulting order is then kept fixed in subsequent forcing extensions, as required by Lemma 7.1.

7.2 Reinterpreting a bounded remainder

Fix $\alpha < \kappa$ and put $M = M_\alpha$. Let $\mathbb{Q}$ be the natural presentation of the designated random or Cohen factor at stage α . For $\eta > \alpha$, factor the portion of the iteration through η as

$$\mathbb{Q} * \dot{\mathbb{T}}_\eta \tag{6}$$

over M . The factorization includes the other coordinates of the stage- α orbit product in $\dot{\mathbb{T}}_\eta$.

Lemma 7.2. *In M , $\mathbb{Q}$ forces that $\dot{\mathbb{T}}_\eta$ belongs to $\mathcal{I}$ and has a dense subset of size at most $\aleph_1$. If $r' \in M_\beta$, $\beta \geq \alpha$, is $\mathbb{Q}$ -generic over M , then $\mathbb{T}_\eta[r']$ belongs to $\mathcal{I}$ in M_β . For every $q \in \mathbb{T}_\eta[r']$, a later stage supplies a $\mathbb{T}_\eta[r']$ -generic filter over $M[r']$ containing q .*

Proof. After the designated factor, the remaining stage- α product is still in $\mathcal{I}$, by Lemmas 2.2 and 2.3. The subsequent iterands are forced to belong to $\mathcal{I}$. Thus the remainder in (6) belongs to $\mathcal{I}$ in the $\mathbb{Q}$ -extension. Its density is at most $\aleph_1$: the bounded initial forcing has such a dense presentation, and its quotient can be represented by the compatible elements of that presentation.

The universe M_β is a c.c.c. forcing extension of M , and $M[r']$ is an intermediate model. By the intermediate-model theorem, M_β is a c.c.c. extension of $M[r']$; a quotient of the relevant c.c.c. Boolean algebra is c.c.c. Since $\mathbb{T}_\eta[r'] \in \mathcal{I}^{M[r']}$, Lemma 2.2 shows that this fixed evaluated order is still in $\mathcal{I}^{M_\beta}$. Normalize a dense presentation and its cone below q . Lemma 5.2 then supplies a later generic filter through q , generic even over a later M_δ , and hence over $M[r']$. $\square$

The evaluation and subsequent forcing with the remainder implement the transfer argument of [5, Claims 6.2–6.3].

7.3 Regularity

Theorem 7.3. *Every boldface Δ_3^1 subset of $\mathbb{R}$ in N is Lebesgue measurable and has the Baire property.*

Proof. Let A be such a set. Choose a real parameter a and two Π_2^1 formulas θ_0 and θ_1 such that in N

$$x \in A \iff \varphi_0(x, a) : \exists z \theta_0(x, z, a), \quad (7)$$

$$x \notin A \iff \varphi_1(x, a) : \exists z \theta_1(x, z, a). \quad (8)$$

Fix a name for a . By the truth lemma, some $p_0 \in G$ forces that these two formulas are complementary for all reals. Choose α so that $p_0 \in G_\alpha$ and the chosen name for a is a $\mathbb{P}_\alpha$ -name. This is possible because conditions have finite outer support and a real has a nice name with countable outer support. Over $M = M_\alpha$, the largest condition of the remaining iteration then forces complementarity.

First use the designated random factor $\mathbb{B}^M$ at stage α . In M , call a positive random condition b *deciding* if there are $i \in \{0, 1\}$, $\eta > \alpha$, and $\mathbb{B}^M$ -names $\dot{q}, \dot{z}$ such that

$$b \Vdash_{\mathbb{B}^M} (\dot{q} \in \dot{\mathbb{T}}_\eta \text{ and } \dot{q} \Vdash_{\dot{\mathbb{T}}_\eta} \theta_i(\dot{r}, \dot{z}, a)). \quad (9)$$

In this notation $\dot{z}$ is a name for a remainder-name for a real, and $\dot{r}$ is the designated random real.

The deciding conditions are dense. Given any positive b , work in the full factorization $\mathbb{B}^M * \dot{\mathbb{T}}_\kappa$ below $(b, 1)$. Complementarity gives a strengthening choosing one side and a name for its real witness. A nice name for this real uses only countably many finite-support conditions, and therefore is already a name at some bounded $\eta < \kappa$. Increase η also to contain the witnessing condition. The bounded condition forces the same Π_2^1 assertion already at that stage: otherwise a bounded strengthening would force its negation, a Σ_2^1 assertion about reals already present, which by Shoenfield absoluteness would remain true in the full extension. This contradicts the chosen full forcing assertion. After strengthening the random coordinate, we obtain (9).

Choose in M a maximal antichain $\mathcal{A}$ of deciding conditions and choose data in (9) for each $b \in \mathcal{A}$. The antichain is countable. Fix Borel representatives for its elements and put

$$B = \bigcup \{b : b \in \mathcal{A}, i_b = 0\}.$$

This is a Borel set coded in M .

Let $r' \in N$ be random over M . Its generic filter meets $\mathcal{A}$ in exactly one element, say b . Evaluate the names attached to b in $M[r']$. By Lemma 7.2, the evaluated remainder has a later generic filter H over $M[r']$ containing $q_b[r']$. The forcing theorem gives

$$M[r'][H] \models \theta_{i_b}(r', \dot{z}_b[r']^H, a).$$

Shoenfield absoluteness carries this to N . Equations (7)–(8) and final complementarity therefore imply

$$r' \in A \iff r' \in B.$$

Null intersections or representative choices for the antichain cause no ambiguity for a real random over M . By Corollary 6.2, the reals which are not random over M form a null set. Thus $A \triangle B$ is null and A is Lebesgue measurable.

For the Baire property, use instead the designated Cohen factor at stage α . Define deciding conditions by the analogue of (9), and choose a maximal antichain of finite strings with the corresponding sides. The union U of the basic open sets assigned side 0 is open. Every real Cohen over M meets this antichain and, by the same evaluated remainder and cone-replay argument, belongs to A exactly when it belongs to U . The exceptional set is meager by Corollary 6.2. Therefore $A \triangle U$ is meager and A has the Baire property. $\square$

8 Failure of definable uniformization

8.1 Homogeneous quotients over coding reals

Lemma 8.1. *Suppose $W[H]$ is a weakly homogeneous forcing extension of W . Every real of $W[H]$ uniquely definable there from parameters in W and ordinal parameters belongs to W .*

Proof. For formulas with ground-model parameters, weak homogeneity implies that the largest condition decides every sentence. Indeed, two conditions forcing opposite answers could be made compatible by an automorphism, while their parameter names remain fixed. If such a definition uniquely specifies a real in $W[H]$, the largest condition consequently forces its existence and uniqueness. For each integer n , it decides the assertion that n belongs to the defined real. The set of integers receiving the positive decision is constructed in W using the forcing relation, and is the specified real. The forcing need not itself be ordinal-definable in W . $\square$

Proposition 8.2. *For every real $p \in N$, there is a real $r \in N$ such that $p \in L[r]$ and N is a weakly homogeneous forcing extension of $L[r]$.*

Proof. Choose α with $p \in M_\alpha$. At stage α , take the designated factor of type 2, which is $\mathbb{A}(A_\alpha)$. Let H be its coordinate generic and r its coding real. Lemma 3.3 gives

$$M_\alpha[H] = L[r].$$

Lemma 4.2 says that the quotient of the full iteration over this exact intermediate extension is weakly homogeneous. Also $p \in M_\alpha \subseteq L[r]$. $\square$

8.2 The counterexample relation

Lemma 8.3. *The relation*

$$R(x, y) \iff y \notin L[x]$$

is lightface Π_2^1 and has total domain in N .

Proof. The standard countable-level characterization of relative constructibility gives

$$y \in L[x] \iff \exists z [\text{WF}(z) \wedge \text{Hier}(z, x, y)].$$

Here z codes a countable constructible hierarchy over x containing y , together with its membership and satisfaction relations. The predicate Hier verifies the successor and limit

steps and the named elements and is arithmetical; WF asserts well-foundedness and is Π_1^1 . Condensation for $L[x]$ supplies a countable such hierarchy whenever $y \in L[x]$. Thus membership in $L[x]$ is Σ_2^1 , uniformly in (x, y) , and its complement is lightface Π_2^1 .

For totality, let $x \in M_\alpha$. A designated Cohen real from a later stage is not in M_α , whereas $L[x] \subseteq M_\alpha$. It therefore witnesses a nonempty R_x . $\square$

Theorem 8.4. *The relation R has no real-ordinal-definable uniformization in N .*

Proof. Suppose F is a uniformization definable in N from a real p and ordinal parameters $\vec{\gamma}$. By Proposition 8.2, choose r such that $p \in L[r]$ and N is a weakly homogeneous forcing extension of $L[r]$.

The relation R has total domain, so there is a unique y with $(r, y) \in F$. This real is uniquely definable in N from $p, r, \vec{\gamma}$, all of which belong to $L[r]$. Lemma 8.1 yields $y \in L[r]$. But $F \subseteq R$ gives $y \notin L[r]$, a contradiction. $\square$

Corollary 8.5. *The lightface Π_2^1 relation R has no ordinal-definable uniformization and no boldface projective uniformization in N .*

Proof. Every ordinal-definable graph is real-ordinal-definable, and every boldface projective graph is definable from a real parameter. Thus both assertions follow from Theorem 8.4. $\square$

Corollary 8.6. *There is no wellorder of $\mathbb{R}$ in N definable from a real and ordinal parameters. In particular, N has no ordinal-definable and no boldface projective wellorder of the reals. Consequently, if ZFC is consistent, then so is*

$$\text{ZFC} + \text{MA}(\mathcal{I}) + \omega_1^L = \omega_1 + \text{“there is no real-ordinal-definable wellorder of } \mathbb{R} \text{.”}$$

Proof. Suppose that $\prec$ is a wellorder of $\mathbb{R}$ definable from a real parameter p and ordinal parameters $\vec{\gamma}$. Since R has total domain, define

$$F(x) = \text{the } \prec\text{-least } y \in \mathbb{R} \text{ such that } y \notin L[x].$$

Then the graph of F is definable from p and $\vec{\gamma}$ and is a uniformization of R , contradicting Theorem 8.4. The special cases follow immediately. Finally, N is a c.c.c. forcing extension of L , so $\omega_1^L = \omega_1^N$, while Proposition 5.3 gives $N \models \text{MA}(\mathcal{I})$. $\square$

Proof of Theorem 1.1. Proposition 5.1 gives the cardinal arithmetic and cardinal preservation. Proposition 5.3 gives $\text{MA}(\mathcal{I})$, Theorem 7.3 gives the two regularity properties, and Lemma 8.3 together with Theorem 8.4 gives the asserted counterexample. The construction is carried out over L without additional hypotheses, so the ordinary forcing interpretation proves the claimed relative consistency from $\text{Con}(\text{ZFC})$. $\square$

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